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Coupling and rotor shaft for molten metal devices

Abstract

A coupling has an opening and a protrusion extending downward from the opening. The protrusion has threads that are preferably positioned outside of the opening. A rotor shaft that connects to the coupling has an internal bore with threads that receives and retains the protrusion, such as by a threaded connection between the two, so the protrusion applies driving force to the shaft.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2035282	12/1935	Schmeller	N/A	N/A
2966345	12/1959	Ciabattari	N/A	N/A
3151565	12/1963	Albertson et al.	N/A	N/A
3258283	12/1965	Winberg	403/379.4	E21B 17/046
4545887	12/1984	Amesen	N/A	N/A
4685822	12/1986	Pelton	403/343	F16D 1/06
4954167	12/1989	Cooper	266/225	B01F 3/04539
4967827	12/1989	Campbell	N/A	N/A
5634770	12/1996	Gilbert	415/216.1	F04D 7/065
5676520	12/1996	Thut	415/121.3	G21C 15/247
5685701	12/1996	Chandler	266/235	F04D 7/065
5744093	12/1997	Davis	N/A	N/A
5758712	12/1997	Pederson	N/A	N/A
6358467	12/2001	Mordue	222/602	B22D 1/005
7273582	12/2006	Mordue	N/A	N/A
7731891	12/2009	Cooper	266/239	F04D 7/065
7841379	12/2009	Evans	N/A	N/A
9074601	12/2014	Thut	N/A	N/A
9494366	12/2015	Thut	N/A	N/A
10428821	12/2018	Cooper	N/A	N/A
10570745	12/2019	Cooper	N/A	N/A
10641270	12/2019	Cooper	N/A	N/A
2011/0140318	12/2010	Reeves et al.	N/A	N/A
2017/0106441	12/2016	Vincent	N/A	N/A
2017/0130298	12/2016	Teranishi et al.	N/A	N/A
2020/0360989	12/2019	Cooper	N/A	N/A
2020/0362865	12/2019	Cooper	N/A	N/A
2020/0363128	12/2019	Cooper	N/A	N/A

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is divisional of and claims priority to U.S. patent application Ser. No. 14/791,137 entitled “Coupling and Rotor Shaft for Molten Metal Devices,” filed on Jul. 2, 2015 which is a non-provisional of and claims priority to U.S. Provisional Application Ser. No. 62/020,332 entitled “Coupling and Rotor Shaft for Molten Metal Devices, filed on Jul. 2, 2014, the contents of each of the foregoing applications are incorporated herein in their entirety for all purposes.

FIELD OF THE INVENTION

(1) The invention relates to couplings and rotor shafts that may be used in various devices, such as pumps, degassers and scrap melters used in molten metal.

BACKGROUND OF THE INVENTION

(2) As used herein, the term “molten metal” means any metal or combination of metals in liquid form, such as aluminum, copper, iron, zinc and alloys thereof. The term “gas” means any gas or combination of gases, including argon, nitrogen, chlorine, fluorine, freon, and helium, that are released into molten metal.

(3) Known molten-metal pumps include a pump base (also called a housing or casing), one or more inlets (an inlet being an opening in the housing to allow molten metal to enter a pump chamber), a pump chamber, which is an open area formed within the housing, and a discharge, which is a channel or conduit of any structure or type communicating with the pump chamber (in an axial pump the chamber and discharge may be the same structure or different areas of the same structure) leading from the pump chamber to an outlet, which is an opening formed in the exterior of the housing through which molten metal exits the casing. An impeller, also called a rotor, is mounted in the pump chamber and is connected to a drive system. The drive system is typically an impeller shaft connected to one end of a drive shaft, the other end of the drive shaft being connected to a motor. Often, the impeller shaft is comprised of graphite, the motor shaft is comprised of steel, and the two are connected by a coupling. As the motor turns the drive shaft, the drive shaft turns the impeller and the impeller pushes molten metal out of the pump chamber, through the discharge, out of the outlet and into the molten metal bath. Most molten metal pumps are gravity fed, wherein gravity forces molten metal through the inlet and into the pump chamber as the impeller pushes molten metal out of the pump chamber.

(4) This application incorporates by reference the portions of the following publications that are not inconsistent with this disclosure: U.S. Pat. No. 4,598,899, issued Jul. 8, 1986, to Paul V. Cooper, U.S. Pat. No. 5,203,681, issued Apr. 20, 1993, to Paul V. Cooper, U.S. Pat. No. 5,308,045, issued May 3, 1994, by Paul V. Cooper, U.S. Pat. No. 5,662,725, issued Sep. 2, 1997, by Paul V. Cooper, U.S. Pat. No. 5,678,807, issued Oct. 21, 1997, by Paul V. Cooper, U.S. Pat. No. 6,027,685, issued Feb. 22, 2000, by Paul V. Cooper, U.S. Pat. No. 6,123,523, issued Sep. 26, 2000, by Paul V. Cooper, U.S. Pat. No. 6,303,074, issued Oct. 16, 2001, by Paul V. Cooper, U.S. Pat. No. 6,689,310, issued Feb. 10, 2004, by Paul V. Cooper, U.S. Pat. No. 6,723,276, issued Apr. 20, 2004, by Paul V. Cooper, U.S. Pat. No. 7,402,276, issued Jul. 22, 2008, by Paul V. Cooper, U.S. Pat. No. 7,507,367, issued Mar. 24, 2009, by Paul V. Cooper, U.S. Pat. No. 7,906,068, issued Mar. 15, 2011, by Paul V. Cooper, U.S. Pat. No. 8,075,837, issued Dec. 13, 2011, by Paul V. Cooper, U.S. Pat. No. 8,110,141, issued Feb. 7, 2012, by Paul V. Cooper, U.S. Pat. No. 8,178,037, issued May 15, 2012, by Paul V. Cooper, U.S. Pat. No. 8,361,379, issued Jan. 29, 2013, by Paul V. Cooper, U.S. Pat. No. 8,366,993,

issued Feb. 5, 2013, by Paul V. Cooper, U.S. Pat. No. 8,409,495, issued Apr. 2, 2013, by Paul V. Cooper, U.S. Pat. No. 8,440,135, issued May 15, 2013, by Paul V. Cooper, U.S. Pat. No. 8,444,911, issued May 21, 2013, by Paul V. Cooper, U.S. Pat. No. 8,475,708, issued Jul. 2, 2013, by Paul V. Cooper, U.S. patent application Ser. No. 12/895,796, filed Sep. 30, 2010, by Paul V. Cooper, U.S. patent application Ser. No. 12/877,988, filed Sep. 8, 2010, by Paul V. Cooper, U.S. patent application Ser. No. 12/853,238, filed Aug. 9, 2010, by Paul V. Cooper, U.S. patent application Ser. No. 12/880,027, filed Sep. 10, 2010, by Paul V. Cooper, U.S. patent application Ser. No. 13/752,312, filed Jan. 28, 2013, by Paul V. Cooper, U.S. patent application Ser. No. 13/756,468, filed Jan. 31, 2013, by Paul V. Cooper, U.S. patent application Ser. No. 13/791,889, filed Mar. 8, 2013, by Paul V. Cooper, U.S. patent application Ser. No. 13/791,952, filed Mar. 9, 2013, by Paul V. Cooper, U.S. patent application Ser. No. 13/841,594, filed Mar. 15, 2013, by Paul V. Cooper, and U.S. patent application Ser. No. 14/027,237, filed Sep. 15, 2013, by Paul V. Cooper.

(5) Three basic types of pumps for pumping molten metal, such as molten aluminum, are utilized: circulation pumps, transfer pumps and gas-release pumps. Circulation pumps are used to circulate the molten metal within a bath, thereby generally equalizing the temperature of the molten metal. Most often, circulation pumps are used in a reverberatory furnace having an external well. The well is usually an extension of the charging well where scrap metal is charged (i.e., added).

(6) Transfer pumps are generally used to transfer molten metal from the one structure to another structure such as a ladle or another furnace.

(7) Gas-release pumps, such as gas-injection pumps, circulate molten metal while introducing a gas into the molten metal. In the purification of molten metals, particularly aluminum, it is frequently desired to remove dissolved gases such as hydrogen, or dissolved metals, such as magnesium. As is known by those skilled in the art, the removing of dissolved gas is known as “degassing” while the removal of magnesium is known as “demagging.” Gas-release pumps may be used for either of these purposes or for any other application for which it is desirable to introduce gas into molten metal.

(8) Gas-release pumps generally include a gas-transfer conduit having a first end that is connected to a gas source and a second end submerged in the molten metal bath. Gas is introduced into the first end and is released from the second end into the molten metal. The gas may be released downstream of the pump chamber into either the pump discharge or a metal-transfer conduit extending from the discharge, or into a stream of molten metal exiting either the discharge or the metal-transfer conduit. Alternatively, gas may be released into the pump chamber or upstream of the pump chamber at a position where molten metal enters the pump chamber.

(9) Molten metal pump casings and rotors often employ a bearing system comprising ceramic rings wherein there are one or more rings on the rotor that align with rings in the pump chamber (such as rings at the inlet and outlet) when the rotor is placed in the pump chamber. The purpose of the bearing system is to reduce damage to the soft, graphite components, particularly the rotor and pump base, during pump operation.

(10) Generally, a degasser (also called a rotary degasser) includes (1) an impeller shaft having a first end, a second end and a passage for transferring gas, (2) an impeller, and (3) a drive source for rotating the impeller shaft and the impeller. The first end of the impeller shaft is connected to the drive source and to a gas source and the second end is connected to the connector of the impeller.

(11) The materials forming the components that contact the molten metal bath should remain relatively stable in the bath. Structural refractory materials, such as graphite or ceramics, that are resistant to disintegration by corrosive attack from the molten metal may be used. As used herein “ceramics” or “ceramic” refers to any oxidized metal (including silicon) or carbon-based material, excluding graphite, capable of being used in the environment of a molten metal bath. “Graphite” means any type of graphite, whether or not chemically treated. Graphite is particularly suitable for being formed into pump components because it is (a) soft and relatively easy to machine, (b) not as brittle as ceramics and less prone to breakage, and (c) less expensive than ceramics.

(12) Generally a scrap melter includes an impeller affixed to an end of a drive shaft, and a drive source attached to the other end of the drive shaft for rotating the shaft and the impeller. The movement of the impeller draws molten metal and scrap metal downward into the molten metal bath in order to melt the scrap. A circulation pump is preferably used in conjunction with the scrap melter to circulate the molten metal in order to maintain a relatively constant temperature within the molten metal.

(13) Numerous rotor shaft to motor shaft couplings are known. A problem with the couplings, however, is that by applying driving force to the rotor shaft the rotor shaft tends to break at the location where the force is being applied. This is typically at the location where the coupling and rotor shaft are in contact, and the broken end of the rotor shaft must often be chiseled out of an opening in the coupling in which it is retained.

SUMMARY OF THE INVENTION

(14) The present invention helps to alleviate the afore-mentioned problems by providing a coupling and rotor shaft for use in a molten metal pump, a molten metal rotary degasser or a molten metal scrap melter. The coupling has a collar with a cavity open at the bottom and a protrusion extending out of the cavity. The protrusion is preferably threaded along at least part of its length, wherein the threaded portion(s) are preferably outside of the cavity.

(15) The rotor shaft preferably has a bore that receives the protrusion. Preferably the bore has internal threads that align with and receive the threads on the protrusion. Most preferably, there are no threads on the inner surface of the coupling opening or the outer surface of the rotor shaft.

(16) With such a structure, if the rotor shaft breaks, it tends to break where the threads of the rotor shaft bore align with the threads of the protrusion, which is below the opening in the coupling. As a consequence, the broken piece of rotor shaft can either be unthreaded or chiseled off of the protrusion, each of which is simpler than chiseling a broken end out of the coupling opening.

(17) The protrusion may have an internal passage for transferring gas therethrough and the rotor shaft passage may extend to the second end (or almost to the second end) to permit gas to be transferred therethrough.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) FIG. 1 is a perspective view of a pump for pumping molten metal, which may include a coupling and rotor shaft according to aspects of the invention.

(2) FIG. 2 shows a rotary degasser, which may include a coupling and rotor shaft according to aspects of the invention.

(3) FIG. 3 shows a scrap melter, which may include a coupling and rotor shaft according to aspects of the invention.

(4) FIG. 4 is a side, cross-sectional view of a coupling that can be used in accordance with aspects of the invention.

(5) FIG. 5 is a partial side view of a rotor shaft that may be used according to aspects of the invention.

(6) FIG. 6 is a side view of a coupling that may be used in accordance with aspects of the invention.

(7) FIG. 7 is a side, perspective view of a coupling that may be used in accordance with aspects of the invention.

(8) FIG. 8 is a partial side view of the lower portion of the coupling shown in in FIGS. 6 and 7.

(9) FIG. 9 is a partial, cross-sectional side view of a rotor/coupling connection in accordance with aspects of the invention.

(10) FIG. 10 is a close-up view of Section G of FIG. 9.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

- (11) For any device described herein, any of the components that contact the molten metal are preferably formed by a material that can withstand the molten metal environment. Preferred materials are oxidation-resistant graphite and ceramics, such as silicon carbide. Oxidation-resistant graphite is most preferred because of its relatively low cost and ease of manufacturing.
- (12) Referring now to the drawing where the purpose is to illustrate and describe different embodiments of the invention, and not to limit same, FIG. 1 shows a molten metal pump **20** in accordance with an aspect of the present invention. Pump **20** is designed for operation in any environment in which molten metal is to be pumped or otherwise conveyed. Pump **20** can be any structure or device for pumping or otherwise conveying molten metal, such as the tangential-discharge pump disclosed in U.S. Letters Pat. No. 5,203,681 to Cooper, or an axial pump having an axial, rather than tangential, discharge. Pump **20** has a pump base **24** submersible in a molten metal bath B. In this embodiment, pump base **24** includes a generally nonvolute pump chamber **26**, such as a cylindrical pump chamber or what has been called a “cut” volute although pump base **24** may have any shape pump chamber suitable of being used, such as a volute-shaped chamber. Chamber **26** may have only one opening, in either its top or bottom, since only one opening is required to introduce molten metal into pump chamber **26**, although chamber **26** may have an opening in both its top and bottom. Generally, pump chamber **24** has two coaxial openings of the same diameter and usually one is blocked by a flow blocking plate mounted on the bottom of, or formed as part of, rotor **100**. Base **24** further includes a tangential discharge **30** (although another type of discharge, such as an axial discharge, may be used) in fluid communication with chamber **26**.
- (13) The invention is not limited to any particular type or configuration of pump base, or of even having a base. A pump, rotary degasser or scrap melter used with the invention could be of any size, design or configuration suitable for utilizing a rotor shaft and coupling according to the invention.
- (14) In the embodiment shown, one or more support posts **34** connect base **24** to a superstructure **36** of pump **20** thus supporting superstructure **36**, although any structure or structures capable of supporting superstructure **36** may be used. Additionally, pump **20** could be constructed so there is no physical connection between the base and the superstructure, wherein the superstructure is independently supported, and/or there is no base. The motor, drive shaft and rotor could be suspended without a superstructure, wherein they are supported, directly or indirectly, to a structure independent of a pump base.
- (15) A motor **40**, which can be any structure, system or device suitable for driving pump **20**, but is preferably an electric, hydraulic or pneumatic motor, is positioned on superstructure **36** and is connected to a drive shaft **42**. Drive shaft **42** can be any structure suitable for rotating the impeller **100**, and preferably comprises a motor shaft (not shown) that connects to rotor shaft **44'** via a coupling **500**, which is described further below.
- (16) A rotor, also called an impeller, **100** is positioned at least partially within pump chamber **26**. Preferred rotor **100** is preferably imperforate, and any type or shape of impeller suitable for use in a molten metal pump, rotary degasser or scrap melter, respectively, may be used to practice aspects of the invention. Rotor **100** has a connecting portion **110** to connect to a second end **44B'** of rotor drive shaft **44'**.
- (17) FIG. 2 shows a preferred gas-release device **200** according to the invention. Device **200** is designed to operate in a molten metal bath B contained within a vessel **1**. Device **200** is preferably a rotary degasser and includes a rotor shaft **44**, an impeller **300** and a drive source (not shown). Device **200** preferably also includes a coupling **500**.
- (18) FIG. 3 shows a scrap melter utilizing a coupling **500** and rotor drive shaft **44'** in accordance with aspects of the invention.
- (19) Turning to FIG. 4, coupling **500** includes a collar **502**, which as shown is circular with a flat top **504**, an annular outer side wall **506**, and a bottom edge **508**. Two slots **505** are preferably

formed on either side of collar **502** so collar **502** may be easily gripped and turned by a wrench. A passage **509** may be formed in collar **502** to permit gas to transfer therethrough. A cavity **510** is formed within coupling **500** in order to receive a first end **38A** of rotor shaft **38**, described further below. In one embodiment the coupling **500** is made of steel, has a height of about 2.5"-3" and an overall width of about 4½"-5½". Cavity **510** has a depth D of about 1-1½", and most preferably about 1¼", a diameter of about 3" next to bottom edge **508**, and an inner wall **512** that slopes inward at between 5° and 10°, and most preferably about 7°, until it terminates at upper cavity wall **514**. Inner wall **512** preferably is not threaded.

(20) A projection **600** extends downward from upper cavity wall **514**, through cavity **510** and projects outward therefrom. Projection **600** may be connected to collar **502** in any suitable manner, such as being threadingly received in an opening (not shown) or welded to collar **502**. In one embodiment, projection **600** is between 0.800 and 1.000 inches in diameter, and preferably about 0.875" in diameter and extends outward from the bottom of cavity **510** (defined by where cavity **510** aligns with bottom edge **508**) by between 2½"-3¾", and preferably 3¼". At the bottom end of projection **600** is preferably a chamfered end **602**, that in the embodiment shown is ¼" long with a 45° chamfer. Projection **600** has an outer surface **604** with threads **606** that are preferably positioned outside of cavity **510**. An upper portion **605** of projection **600** is positioned in cavity **510**. As shown, the threads **606** preferably begin at a position about ½" beneath the lower portion of cavity **510**, extend for about 1¼", and are 1" ACME threads, although any suitable length and type of thread may be used. Projection **600** may have an internal passage **509** therein for the passage of gas.

(21) Coupling **500** may also include one extended metal portion **650** above coupling Section **502**. The extended length of portion **650** reduces the cost of new and replacement rotor shafts (because they are shorter) and portion **650** is preferably between 4.5" and 12" in length. A top plate **660** mates with a known structure, such as a motor shaft with a rotary union, for driving the coupling **500** and injecting gas into it. Apertures **662** permit fasteners (not shown) to pass through and connect to a suitable driving structure, which is known in the art.

(22) As is illustrated in FIGS. 2 and 5, rotor shaft **38** has a first end **38A**, a second end **38B**, an outer surface **706** (that is preferably annular) and an inner passage **708** (which is optional depending upon the application) for transferring gas. Second end **38B** preferably has a structure, such as a threaded end, for connecting to an impeller, although any suitable connection may be used. Shaft **38** may be a unitary structure or may be a plurality of pieces connected together.

(23) First end **38A** is dimensioned to receive projection **600** and to fit into cavity **510**. First end **38A** as shown has a chamfered end **700** that is received in cavity **510** so that top surface **708** is against surface **514** when the rotor shaft **44** and coupling **500** are assembled. First end **38A** also has an opening **712** leading to inner passage **708**, which includes threads **714** that threadingly engage threads **606**. In this embodiment, inner passage **708** extends through, or essentially through, rotor shaft **38** to transfer gas to second end **38B** so that gas may be released into molten metal in any suitable manner.

(24) Rotor shaft **38** is connected to coupling **500** by aligning projection **600** with opening **712** and pushing projection into passage **708** until threads **606** meet threads **714**, and then screwing rotor shaft **38** onto projection **600**. Most preferably the respective threaded portions **606** and **714** are tightened in the same direction as the rotor shaft rotates during operation so that they do not loosen during operation.

(25) Having thus described different embodiments of the invention, other variations and embodiments that do not depart from the spirit of the invention will become apparent to those skilled in the art. The scope of the present invention is thus not limited to any particular embodiment, but is instead set forth in the appended claims and the legal equivalents thereof. Unless expressly stated in the written description or claims, the steps of any method recited in the claims may be performed in any order capable of yielding the desired result.

Claims

1. A rotor shaft configured to connect to a coupling, the rotor shaft comprising: a first end configured to connect to the coupling, the first end comprising (a) an outer surface having a chamfer that is configured to be received in a cavity of the coupling, and (b) an internal bore, wherein there are no threads in the internal bore at the position of the chamfer and there are threads in the internal bore only beneath the position of the chamfer, wherein the threads are configured to receive threads of a protrusion extending from an opening of the coupling in order to be threadingly connected to the protrusion.
 2. The rotor shaft of claim 1 that is comprised of one or more of the group consisting of: graphite and ceramic.
 3. The rotor shaft of claim 1 that has an outer diameter of 3" and the first end is chamfered to reduce the outer diameter to 2¾" over a 1" to 1.5" length of the rotor shaft.
 4. The rotor shaft of claim 1, wherein the bore has a first end and threads begin at least 1¼" past the first end of the bore.
 5. The rotor shaft of claim 4, wherein the threads are square, 1", ACME threads.
 6. The rotor shaft of claim 1, wherein the bore extends through the rotor shaft and is configured for the passage of gas to a second end of the rotor shaft.
 7. The rotor shaft of claim 1 that has 1" ACME square threads.
 8. The rotor shaft of claim 1 that has a second end and an exterior surface, wherein the exterior surface is threaded at the second end.
 9. The rotor shaft of claim 1, wherein the outer surface of the first end is chamfered at a 45° angle.
 10. The rotor shaft of claim 1, wherein the chamfered outer surface of the first end is 1"-1.5" in length.
 11. The rotor shaft of claim 1, wherein the outer surface of the first end is chamfered at a 7° angle.
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